From Data to Delivery: Predicting Pregnancy Risk with a Boosted Decision Tree

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Introduction and Biological Background

With an estimated 213 million pregnancies occurring around the world each year, prioritizing maternal health is crucial in ensuring safe and successful pregnancies. The National Academies of Sciences et al. defines a 'high risk pregnancy' as a "*situation in which the pregnant woman, fetus, or both have an increased likelihood or odds of a pregnancy complication, adverse event, or poor outcomes occurring during or after the pregnancy or birth.*"[2]. Holness outlines that The Centers of Disease Control and Prevention estimate that in the United States, 65,000 women have high risk pregnancies, annually[3]. The Maternal Mortality Review Committees stated that 80% of pregnancy related deaths in the United States were preventable[4]. Furthermore, The World Health Organization states that in 2020, 95% of maternal deaths occurred in low and lower-middle income countries, specifically in Sub-Saharan Africa and Southern Asia[5]. These sources outline the demand for the increase in the quality prenatal services available, especially to those who come from a lower socioeconomic background.

Ahmed et al conducted a series of surveys at hospitals and maternity clinics in Dhaka and Khulna, Bangladesh. Through the use of Radio-Frequency Identification sensors, data was collected and recorded on pregnant mothers. Ahmed et al assessed which factors lead to a high risk pregnancy[6]. The collected data contains 1,014 observations and 7 features, 'Age,' 'SystolicBP,' 'DiastolicBP,' 'BS,' 'BodyTemp,' 'HeartRate,' and 'Risk Level':

- Age: The number of years an individual has been alive.
- SystolicBP (Systolic Blood Pressure): The upper value of a blood pressure reading, measured in millimeters of mercury (mmHg). A healthy individual's Systolic Blood Pressure typically ranges between 90-120 mmHg.
- DiastolicBP (Diastolic Blood Pressure): The lower value of a blood pressure reading, measured in millimeters of mercury (mmHg). A healthy individual's Diastolic Blood Pressure typically ranges between 60-80 mmHg.
- BS (Blood Sugar): The amount of glucose in the blood stream, measured in molar concentration (mmol/L). A healthy individual's blood glucose levels are less than or equal to 7.8 mmol/L.
- BodyTemp (Body Temperature): The internal temperature of an individual, measured in Fahrenheit degrees. The typical body temperature for a healthy individual is 98.6 degrees Fahrenheit.
- HeartRate (Resting Heart Rate): The amount of times an individual's heart beats in one minute. The typical resting heart rate of a healthy adult is between 60 and 100 beats per minute.
- Risk Level: The predicted risk intensity level during pregnancy. The Risk Level classifies patients into three categories: 'low risk,' 'mid risk,' or 'high risk'.

If a thorough evaluation on the data can be performed, and a model that correctly classifies the observations as 'low risk,' 'mid risk,' or 'high risk' can be made, then alternative methods to encourage patients to seek prenatal care can be created.

Model Development

The data were prepared by loading them into the programming environment as a *Pandas' DataFrame*. The proposed predictor, "RiskLevel", was removed from the dataframe and set equal to the variable y. The remaining dataframe was set equal to the variable X. Using scikit-learn's *train_test_split* function, the dataframe was divided into:

$$X_{\text{train}} = \{x_i \in X \mid i \in \text{train indices}\}$$
$$y_{\text{train}} = \{y_i \in y \mid i \in \text{train indices}\}$$
$$X_{\text{test}} = \{x_i \in X \mid i \in \text{test indices}\}$$

$$y_{\text{test}} = \{y_i \in y \mid i \in \text{test indices}\}$$
$$|X_{train}| = 0.8 * N$$
$$|y_{train}| = 0.8 * N$$
$$|X_{test}| = 0.2 * N$$
$$|y_{test}| = 0.2 * N$$
where N is the total number of data points in X and y.

Therefore, 80% of the X and y data were put into a train set, and the remaining data (20%) into a test set. This process yielded four sets, $X_{train}, y_{train}, X_{test}$, and y_{test}, as stated above. The *random_state* argument was used to ensure that if the coding cell was run again, the split would remain the same. Finally, the class names in y were changed from, 'low risk,' 'mid risk,' and 'high risk' to 0, 1, and 2, respectively.

A benchmark logistic regression (LR) model was applied to the dataset to assess whether there was a relationship between the features and the predicted values. First, scikit-learn's *LogisticRegression* function was loaded in. Then, the LR model was fit to the X_{train} and y_{train} data. Finally, the X_{test} data was fed into the model and was used to make predictions, $\hat{y}$. To obtain an accuracy score, y_{test} and $\hat{y}$ were compared using scikit-learn's *accuracy_score* function. The accuracy score of the LR model was interpreted as follows: if

$$accuracy_{BenchmarkLR} \geq 0.57,$$

then, a relationship between X and y does exist, and it is possible to create a more complex and robust model to make predictions.

After assessing several baseline classification methods and parameter tuning strategies, ensemble trees and boosting were the models selected. Examples of the baseline methods and the parameter tuning strategies explored include K-nearest-neighbors, support vector machines, random forests, bagging, lasso regression, and ridge regression.

A baseline tree was created using scikit-learn's *DecisionTreeClassifier* function. The model was loaded into the coding environment, and then fit to the X_{train} and y_{train} sets. The *random_state* argument was used to ensure that if the coding cell was run again, the model would remain the same. The X_{test} data was fed into the model and was used to make predictions, $\hat{y}$. To obtain an accuracy score, y_{test} and $\hat{y}$ were compared using scikit-learn's *accuracy_score* function.

Additionally, a tree depth and maximum nodes finder was made using scikit-learn's *GridSearchCV* function. 10-fold cross-validation splits were set, using scikit-learn's *KFold* function. *GridSearchCV* was loaded into the coding environment and the DecisionTreeClassifier model was passed as an argument during the construction of the GridSearchCV model. The GridSearchCV model was fit to the X_{train} and y_{train} sets, and GridSearch's *best_params_* and *best_estimators_* methods were used to find the tree depth and maximum nodes. Scikit-learn's *cost_complexity_pruning_path* function was used to prune the tree and find the maximum nodes. The same steps used to find the tree depth were used to prune the tree; *cost_complexity_pruning_path* was loaded into the environment, the Tree model was passed as an argument into the construction of the pruning model, and the data were fit.

To construct the boosted tree model, the *GradientBoostingClassifier* function was used. *GradientBoostingClassifier* was introduced to the environment, fit to the X_{train} and y_{train} sets, and then made predictions on X_{test}. Next, a custom hyperparameter finder was used to identify the optimal learning rate and number of estimators. The following sets of hyperparameters were tested:

$$learn_rates = [0.001, 0.01, 0.1, 1]$$
$$estimators = [900, 1000, 1250, 1500, 1750, 2000]$$

The boosting model was then recreated with the optimal learn rate and estimator size. The tree depth and maximum node, found from *best_params_* and *best_estimators_*, were applied to the Boosted models.

To visualize the performance of the model, a confusion matrix of the test data was created using scikit-learn's *confusion_matrix* function, and seaborn's *sns* class and *heatmap* method.

To asses the validity of the model, 10-fold cross validation was conducted using scikit-learn's *KFold* function. The model was introduced to the environment with the arguments *shuffle* set to true and *random_state*. Additionally, along with an accuracy score, the precision, recall, and F1 scores of the model were calculated using scikit-learn's *precision_score*, *recall_score*, and *f1_score* functions. The Type I error and Type II error were also measured.

Results

Figure 1 illustrates our Logistic Regression model made. Note that for visualization purposes, one feature ('Blood Sugar') was selected, and the other features were left constant (averaged). See figure 1 (page 3):

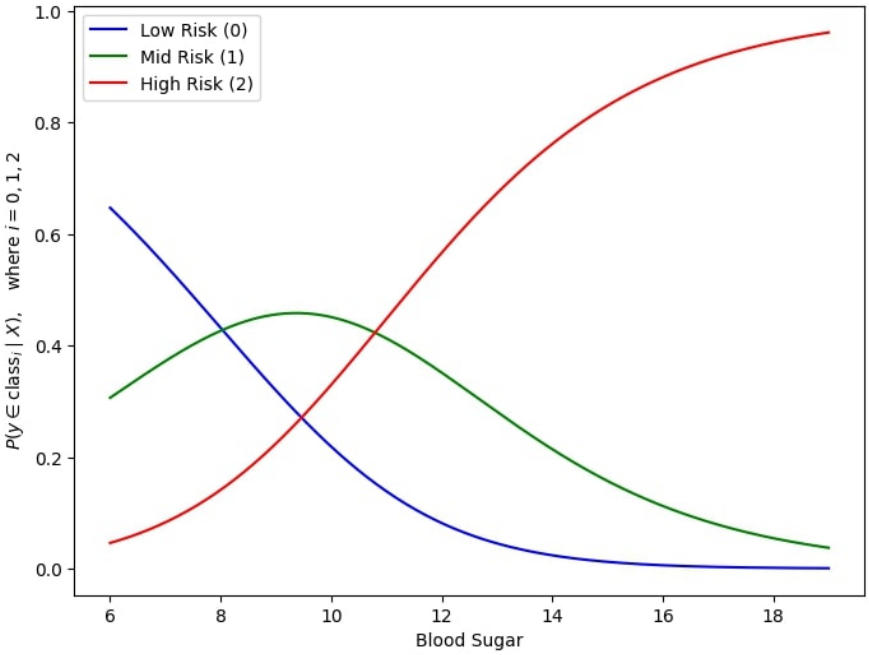

Figure 1: Plot of Logistic Regression Model, with $X = X_0 =$ Blood Sugar and where $X_1, ..., X_N$ are Averaged

Our Benchmark Logistic Regression model yielded an accuracy score of

$$accuracy_{BenchmarkLogisticRegression} = 0.6502463054187192$$

Next, the Baseline Decision Tree was made. See figure 2:

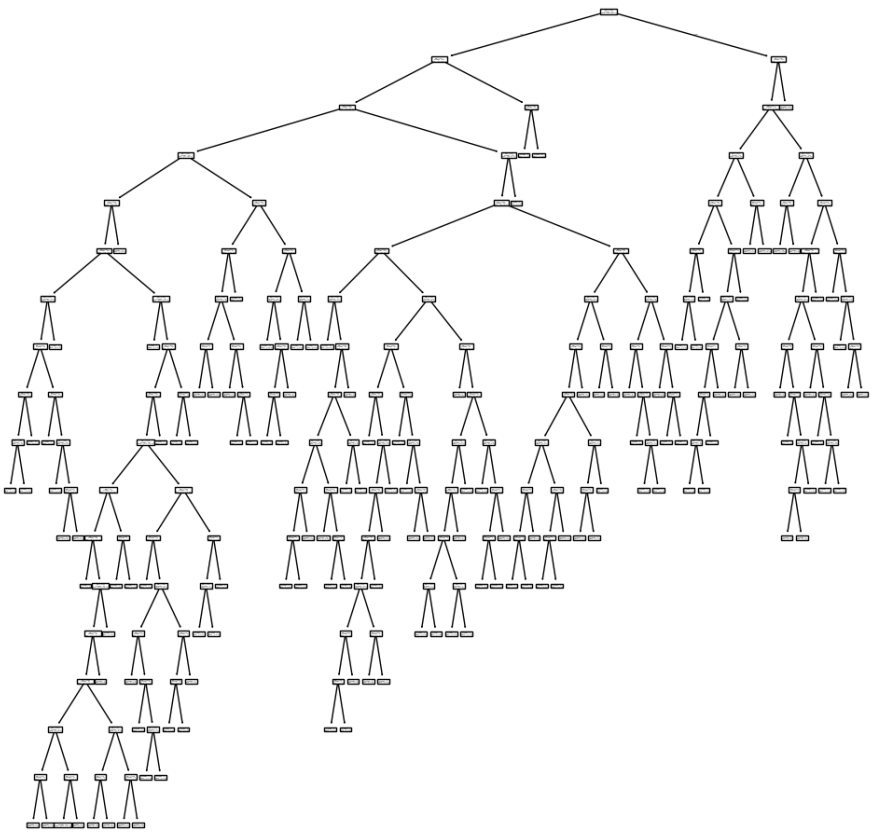

Figure 2: Baseline Ensemble Tree, with a Maximum Node count of 293and a Maximum Depth of 17

The Tree depth and maximum nodes finder assessed that;

$$Best\ Max\ Depth = 17$$
$$Max\ Nodes = 293$$

The accuracy of the Baseline Decision Tree, with these hyperparameters, was found to be:

$$accuracy_{BaselineDecisionTree} = 0.8226600985221675$$

The first Boosted model yielded an accuracy score of:

$$accuracy_{UntunedHyperparameterBoostedTree} = 0.8177339901477833$$

A custom hyperparamter finder was used to detect the best learning rate and estimator amount. The hyperparameters were found to be;

$$n_estimators = 1500$$
$$learn_rate = 0.01$$

Another Boosted model was created, taking into account the optimal hyperparameters. The accuracy of the second Boosted model was found to be;

$$accuracy_{TunedHyperparametersBoostedTree} = 0.8325123152709359$$

Additionally, the Confusion Matrix was made. See figure 3;

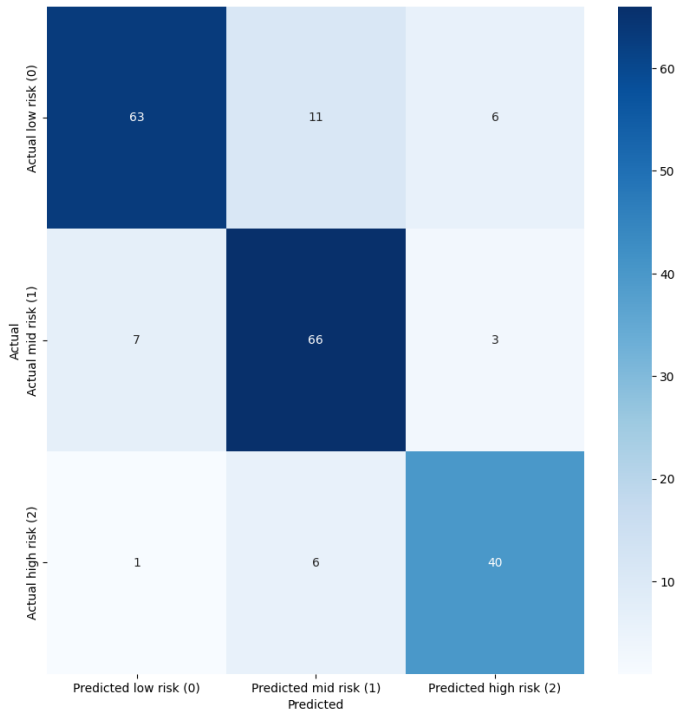

Figure 3: Confusion Matrix of Boosted Tree Model Predictions

After performing 10-fold cross validation, the following metrics were found:

$$10 - Fold\ Cross\ Validation\ Scores =$$
$$[0.81372549, 0.81372549, 0.83333333, 0.89215686, 0.84158416, 0.87128713, 0.91089109, 0.88118812, 0.83168317, 0.89108911]$$
$$Accuracy_{MeanOver10Folds} = 0.8580663948747815$$
$$Standard\ Deviation_{AccuracyOver10Folds} = 0.03356218800206466$$

Finally, precision, recall, F1, Type I and Type II error were calculated. See below;

$$precision_{TunedHyperparametersBoostedTree} = 0.8320681931309108$$
$$recall_{TunedHyperparametersBoostedTreeClass_0} = 0.7875000000000000$$
$$recall_{TunedHyperparametersBoostedTreeClass_1} = 0.8684210500000000$$
$$recall_{TunedHyperparametersBoostedTreeClass_2} = 0.8510638300000000$$
$$F1_{TunedHyperparametersBoostedTree} = 0.8326000000000000$$
$$Test\ I\ Error_{TunedHyperparametersBoostedTree} = 0.16748768472906406$$
$$Test\ II\ Error_{TunedHyperparametersBoostedTree} = 0.16793180686908915$$

To help with model interpretation, the level of importance of each predictor was found. See figure 4:

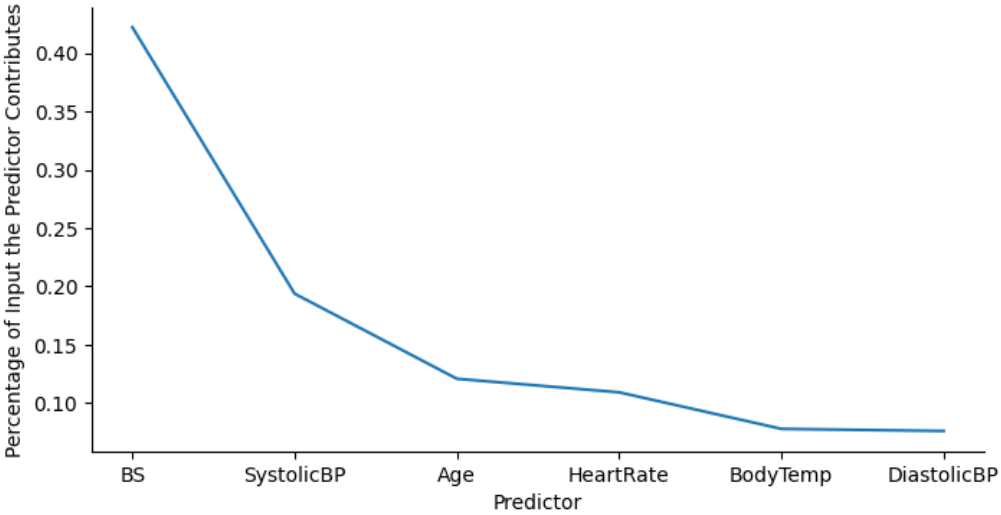

Figure 4: Importance of Predictors in the Boosted Tree Model

Discussion

This analysis aimed to develop a model capable of classifying pregnancies into three categories— 'low risk,' 'mid risk,' or 'high risk' —based on commonly observed factors during pregnancy. First, our Benchmark Logistic Regression (LR) model was made. The accuracy score of the LR model was objectively not optimal. However, the conclusion that the Model was not merely guessing could be drawn, as the obtained accuracy of 65% exceeded the threshold set of 57%. Therefore, a more robust and complex model could be created to make accurate predictions.

Next, our pruned Baseline Decision Tree was made, and the optimal node count and tree depth was found. Typically, a tree of 293 nodes and of depth 17 would lead to the conclusion that our Tree is overfit, but because the data used was complex and highly nonlinear, a complex decision tree was needed to make correct predictions. Both the Untuned Boosted Tree and the Tuned Boosted Tree were made, and yielded the accuracy scores of 81.77% and 83.25%, respectively. Although the difference is small, it was deemed significant, as the additional 2% translates to accurately categorizing 20 more pregnancies.

10-Fold cross validation displayed accuracy scores ranging from 70% to 92%. Several conclusions could be drawn from this. First, the data could be noisy or contain lots of outliers in some folds. Second, some classes could be overrepresented in some folds, while some are underrepresented in others, leading to the overfitting or underfitting of the Model.

Our Confusion Matrix (CM) of the test data was created. The CM shows that the majority of misclassifications made were because a low risk pregnancy was classified as a mid risk pregnancy (roughly 32% of misclassifications). Although this was not desired, the real world results of this kind of missort would not be as detrimental as the missort of an actual high risk pregnancy.

Additional metrics were found to assess the validity of the Model. Note that the recall scores of the classes vary slightly, with low risk's recall score being 78%, and mid risk and high risk's recall scores being 85% and 86%, respectively. This could be because the Model is failing to learn 'low risk' as well as it is learning 'mid' and 'high risk'. Again, although a misclassification was never desired, the real world results of a low risk pregnancy missort are not as harmful as the missort of a mid or high risk pregnancy. Note that the Test I and Test II errors were both high (16.7%). Further hyperparameter tuning should be investigated to minimize these scores.

As shown in figure 4, 'Blood Sugar' was consistently the most significant variable in predicting the risk level of pregnant women; in predicting, it contributed 42% of the input. Blood sugar levels play a large role in diabetes, which is often outlined as a major contributing factor to pregnancies being high risk. The model selecting 'Blood Sugar' to be an important variable, while literature also notes its importance, supports the biological validity of the Boosted Tree Model. Although 'Systolic Blood Pressure' was the second most important variable in making predictions, its importance was only 19%, which is a significant drop from the importance of 'Blood Sugar'. The third most important factor, 'Age', is commonly noted as a major factor in classifying high risk pregnancies. However, our Boosted model showed that Age only contributes to 12% of the input when making predictions. This is far lower than what would be expected based on literature. In her contribution to the book,"Female Health Across the Lifespan", Holness outlines four main categories that high risk pregnancy factors fall into, with 'Age' being its own category[3]. These conclusions support the idea that health professionals should further investigate how blood

sugar levels contribute to high risk pregnancies, and they should potentially place more emphasis on diabetes when diagnosing the risk level of a pregnancy. The remaining predictors', "HeartRate", "BodyTemp", and "DiastolicBP", importance was less than or equal to 10%, and so they were not assessed in this write-up.

Conclusion

In summary, this analysis demonstrated the development and evaluation of a Boosted Tree model for classifying pregnancy risk. Starting with a Benchmark Logistic Regression model, the data was confirmed to possess the potential for predictive accuracy, despite its initial limitations. The transition to decision tree and boosted tree models highlighted the complexity of the dataset, with the tuned boosted tree achieving a cross validated accuracy of 85.81%. Cross-validation underscored the variability in the dataset, while a confusion matrix revealed meaningful insights into misclassifications and their practical implications. The study also identified "Blood Sugar" as the most significant predictor, aligning with existing literature on the role of diabetes in high-risk pregnancies. These findings suggest that further refinement of the model, such as continuing to fine tune hyperparameters, and placing an emphasis on managing blood sugar could enhance clinical decision-making and improve outcomes in maternal health.

A model like this could be transformed into an interactive online quiz, providing an accessible tool for pregnant individuals to evaluate their pregnancy risk. By making such an evaluation available online, access to prenatal care could be significantly expanded, especially for those who might not otherwise seek it. Example questions might include: 'How old are you?', 'Do you have diabetes?', and 'On a scale from 1 to 10, what is your daily physical activity level?'.

Although the latter two questions do not directly align with the model's predictors, observations could be inferred from the provided responses. For instance, if a person has diabetes, their estimated 'Blood Sugar' observation might be ≥ 11.1 (mmol/L), consistent with the elevated glucose levels typical for individuals with diabetes. Similarly, if someone reports a physical activity level of ≥ 7, their estimated 'Heart Rate' could be ≤ 80 (bpm), reflecting the inverse relationship between activity level and resting heart rate.

Upon completing the quiz, individuals classified as high risk would be encouraged to seek prenatal care. The quiz could include links to local obstetricians and other pregnancy-related support services to guide users towards appropriate resources.

The overarching aim of the Boosted Model would be twofold: to motivate expecting parents to pursue prenatal care, especially those who might not have otherwise, and to educate them about their health options and available support systems throughout pregnancy.

References

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